

Student Success Report

Portuguese Student Success

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Introduction

Student retention has become a growing concern for higher education institutions worldwide. Dropout not only impacts institutional performance metrics but also has long-term financial and psychological consequences for the students themselves. This project leverages data analytics to identify students who are at risk of dropping out, enabling the development of targeted early intervention strategies.

The dataset used in this analysis comes from a public Portuguese higher education institution and includes academic, demographic, and socioeconomic information for approximately 4,424 students across different semesters. It contains both categorical and continuous variables, such as marital status, admission grade, tuition payment status, and academic performance across the first and second semesters.

The primary objective of this project is to develop a predictive model that classifies students into two categories: Graduate or Dropout. To guide this analysis, the project focuses on two overarching business questions:

- Which student dimensions are most predictive of dropout?
- What types of student profiles are at the highest risk of dropping out?

To answer these questions, we applied two predictive modeling techniques (Random Forest and Logistic Regression). These approaches helped quantify dropout risk, identify the most influential variables, and uncover distinct high-risk student profiles.

This report outlines the data preparation process, modeling techniques, evaluation metrics, and interpretation of results, followed by a discussion of actionable insights and key limitations.

Data Description

We obtained the dataset from Kaggle, however it was initially gathered and cleaned by Realinho, Machado, Baptista, and Martins (2022) from the Polytechnic Institute of Portalegre. The original authors collected data from multiple sources. According to Realinho et al. (2022) they were from the institution's Academic Management System (AMS), the Support System for Teaching Activity (PAE), the General Directorate of Higher Education (DGES) admission data and the Contemporary Portugal Database (PORDATA) for macroeconomic indicators. After collecting it, they handled all preprocessing for outliers and missing values, providing a clean dataset for us to use. This allowed us a head start on our own data preparation, like transforming variables for skewed distributions and outliers.

In this dataset there are 4,424 students (observations) who are enrolled in several undergraduate programs between the years of 2008/2009 and 2018/2019. There are 37 variables (continuous and categorical). The target variable is students' academic status, whether they are a graduate dropout or are enrolled. 49.9% of students are graduates (2,209), 32.1% are dropouts (1,421) and 17.9% are enrolled (794). There are academic variables such as Previous qualification grade, Admission grade, Application mode and order, Course program and attendance type (daytime/evening), Previous qualification type. There are academic performance variables such as Curricular units grades (1st and 2nd semester), Curricular units credited/enrolled/evaluated/approved (1st and 2nd semester), Grade point averages by semester. There are also demographic variables such as Age at enrollment, Gender, Marital status, Nationality, Displaced status. There are socioeconomic variables such as Mother's and father's qualifications, Mother's and father's occupations,

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Tuition fees up to date, Scholarship holder status, and Debtor status. Finally, there are also macroeconomic variables such as Unemployment rate, Inflation rate and GDP.

Since enrolled students' outcomes are unknown, they were excluded, resulting in a dataset of 3630 students, creating a binary classification route. In the end there were no missing values, and the sample size still remained significantly large for modeling and analyzing.

Models and Methodologies

To answer our two business questions, two different supervised machine models have been used. A logistic regression model was developed to identify the most significant predictors of student graduation or dropout. It quantified the relationship between our target variable and outcome variables. The coefficients highlighted the most important factors of our prediction.

A random forest classifier was used to create student profiles. It creates and gathers different decision trees to improve model predictions while ranking important features. It is extremely useful for identifying recurring characteristics and creating student profiles.

Single Dimensional Modeling

1. Academic Program:

The program status of students in Portugal was one of the key areas we looked at to assess what factors impacted student dropout rates. Across this dimension, we looked at students in daytime and evening classes and students with educational special needs. To model the daytime and evening students, we split the data into students in daytime classes and students in evening classes and created models for each of those students. For students

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in daytime classes, they were slightly less likely to drop out, with a dropout rate of only 30.8% compared to students in evening classes who had a dropout rate of 42.9%.

Both the logistic regression and random forest models for daytime and evening students had high accuracy and AUC-ROC scores. For daytime students, important features included Curricular units 2nd and 1st sem (approved), Curricular units 1st sem (grade), Tuition fees up to date, and Age at enrollment. This highlights the importance of first-semester performance in dropout risk. Students who perform well in the first semester are less likely to drop out. For evening students, important features include Displaced, Father's qualification, Application order, Debtor, and Scholarship holder. These features indicate that evening students might need more financial support to succeed compared with daytime students. This would make sense because students in evening classes might be taking evening classes so they can work during the day highlighting the need for increased financial support.

We also looked at students in the educational special needs program to assess their dropout risk. These students were harder to predict because there weren't very many of them. The best model was a logistic regression model with an accuracy of only 78%. The important features included Curricular units 2nd sem (approved), both curricular units 1st and 2nd sem (grade), and admission grade. These features highlight the importance of academic success and the support of students with educational special needs. Students with educational special needs who do well are less likely to drop out than those who are struggling.

2. Demographics

Demographic data provides insight into students' characteristics. Students come from a variety of backgrounds and understanding their different characteristics might reveal different factors that affect their risk of dropping out.

A full logistic model with gender, marital status and displaced status had both an accuracy of 65% and AUC-ROC of 65%. It indicates that demographics play a somewhat significant role in determining if students will drop out or not. The highest predictor or coefficient was gender, following it was both the marital status and displaced status of students.

With gender, specifically males, being the highest predictor, it points out that certain programs might or might not be of interest to male students. It might also indicate that male students tend to think less of higher education in comparison to females. Similarly, single students and displaced students were the most likely to graduate. It could be interpreted as students who are married/widowed/divorced do not have as much free time as single students to focus on studying. Special programs could be provided to offer flexible classes for those with prior commitments. As for displaced students, it is hard to understand why they graduate more than those who are not displaced, but a possible reason would be due to prior hardships that act as a driving force for them to accelerate in their studies.

Our random forest model also provided similar metrics and results to the logistic regression model. Even though the nature of a random forest and a logistic regression

greatly differ, they both indicate the importance of gender. Further research could be conducted to understand why this variable affects drop out.

3. Grade Progression Measures

The semester grades showed a negatively skewed distribution, meaning that most students did really well. To fix this, we used reflect-log transformations, which flipped the scale and made the data more normal. Now, higher transformed values actually mean poorer academic performance. The Portuguese education system grades on a scale of 0-20. The second-semester grades came out as the most important out of all predictor in all models. Variable importance procedures revealed 2nd semester grades a most predictive. The grade was split into 3 levels. Poor performers with grades below 7-8 have a 69% dropout risk, average performers with grades between 8-14 drop out at about 18%, and good performers, with grades above 14, have an 11% dropout risk.

In the modeling stage for both logistic regression and random forest as students performance improved predictive accuracy decreased. Poor performers showed 88.3% accuracy and 94.5% AUC, average performers (91.2% accuracy and 84.2% AUC), finally good performers showed 90.4% accuracy and 79.6% AUC. Among the high achievers, it became more difficult to assess the dropout risk. Finally, we applied recursive feature elimination with cross-validation (RFECV) and came up with 20 features for poor performers and 15 features for average/good performers while maintaining strong performance. Poor performers had a final accuracy of 85.80% after applying RFECV. Average performers had a 90% final accuracy, and good performers had a 89.81% final accuracy. Poor performer's 3 most important features. Curricular units 2nd semester approved (97.30 MeanDecreaseGini), curricular units 1st semester approved (54.36

MeanDecreaseGini), and curricular units 2nd semester grade (43.11 MeanDecreaseGini). To benefit academic institutions, this analysis suggests that poor performers need additional support and immediate intervention. Good performers need more variables to assess their reasons for leaving school since we have less predictive accuracy in that level. The limitations in the approach of this dimension were careful interpretation of the transformed grade variables, since they flipped the scale. Also there is a strong correlation between semester grades 1st and 2nd semester ($r = 0.828$) however this is not a multicollinearity issue rather it shows how natural academic progress flows from one semester to the next. The most important finding in this dimension was that most recent grades matter the most, rather than high school grades, or admission grades, when predicting dropout risk.

4. Financial Status

The financial status of Portuguese students emerged as a critical dimension influencing dropout risk. To assess this relationship, we built two classification models, as mentioned above – Random Forest and Multinomial Logistic Regression – using financial variables, including debtor, tuition fee up to date, and scholarship holder.

Both models achieved an accuracy of 60.62%, indicating moderate predictive power when using financial features alone. To further assess performance, we calculated the Area Under the Curve (AUC), with Random Forest's AUC achieving 65.54% and Logistic Regression at 72.14%. These AUC statistics suggest that the models can distinguish between dropout and graduate outcomes better than random guessing.

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From Random Forest model, among the financial variables, tuition fee up to date emerged as the most important predictor, with a Gini importance score of 132.92, followed by scholarship holder (48.87), then debtor (24.64). Students with tuition fees paid up to date were significantly less likely to drop out.

Looking at multinomial logistic regression, all three variables were statistically significant predictors of dropout:

- tuition_fees_up_to_date: Estimate = 3.13, $p < 0.001$
- debtor: Estimate = -0.98, $p < 0.001$
- scholarship_holder: Estimate = 1.64, $p < 0.001$

These coefficients reflect the directional impact and strength of each predictor in relation to dropout status.

Overall, the financial dimension proved to be a critical predictor of student dropout risk. Among the three key variables analyzed, tuition fees up to date consistently emerged as the strongest indicator across both models—students who were current on tuition payments were 3.1 times more likely to persist in their studies, highlighting the importance of financial stability in academic success. Additionally, students identified as debtors were more than twice as likely to drop out compared to their non-debtor peers, reinforcing the idea that financial strain plays a major role in attrition. Finally, being a scholarship holder was associated with significantly higher retention rates, suggesting that financial aid helps reduce dropout risk by alleviating students' economic pressures. These findings emphasize the value of targeted financial support programs as a strategic intervention to improve student outcomes.

Validation

Model validation is important to protect against overfitting, making our models more efficient, and to improve their performance. To validate our models, we used a combination of recursive feature elimination and cross-validation. Cross-validation is a technique that works by splitting the data into k folds and training the data on $k-1$ of those folds and testing on the last fold. We split our data into five folds and ran through 5 iterations to test our models on each of the five folds. We also combined this with recursive feature elimination to reduce the number of features in our model. Recursive feature elimination is a technique that iteratively removes features from the model based on importance until the desired number of features is reached. Our dataset had 37 features, which made it difficult to improve the accuracy of our models without losing important information. We found that combining cross-validation with recursive feature elimination was the best way to find the model with the optimal set of variables. After implementing recursive feature elimination and cross-validation on both the full logistic regression and random forest both models had improvement. Our overall logistic regression model improved from an accuracy of 90.9% to 91.12%, and the number of features decreased to 20 after the implementation of model validation, highlighting increased performance and efficiency. After implementation of cross-validation and recursive feature elimination for our overall random forest model, the accuracy didn't improve very much, but the number of features decreased to 22 with the same accuracy, showing our validation improved our model efficiency.

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Analysis of Results

Looking at our overall model results, the best model was a Logistic Regression model with 20 features and an accuracy of 91.12%, indicating excellent performance. The most important features included Tuition Fees Up to Date, Debtor, Curricular units 2nd and 1st sem (approved), and Scholarship Holder. Tuition Fees up to Date had an importance metric of 2.15, which was the highest importance by far. This shows the importance of the financial status of students in determining dropout. If students are doing well financially and don't have to worry about paying for school, they are more likely to succeed in school.

This is also in line with our single-dimensional modeling, as the financial assistance dimension was a key indicator in predicting student dropouts. We also found that the scholarship holder feature was positively correlated with student retention, highlighting how institutions can help students find success.

Conclusion

Overall, this project aims to develop a predictive model that identifies students at risk of dropping out, enabling institutions to implement targeted interventions that enhance retention and academic resources. Academic performance, particularly second-semester grades, emerged as the strongest predictor of dropout risk in grade progression dimension. Financial factors were also highly influential, as mentioned above, students who were current on tuition and are scholarship holder are more likely to graduate, compared to students who identify as debtors, highlighting the importance of financial aid in promoting persistence. Besides that, demographic and program-related insights revealed that students in evening programs or with special needs were at higher risk, indicating the

need for more personalized academic support. Conversely, displaced and single students showed stronger retention, suggesting motivational or contextual resilience.

Our final logistic regression model achieved 91.2% accuracy and an AUC of 90.13%, outperforming the random forest alternative. It used 20 key features, including tuition status, debtor status, and academic progress metrics, to balance accuracy with interpretability. In response to these insights, we recommend several targeted actions: proactive financial outreach, flexible tuition payment plans, and expanded scholarship opportunities for financially at-risk students; early academic interventions for underperforming students; and enhanced support systems for vulnerable populations such as evening students and those with special needs. These recommendations are intended to be scalable, cost-effective, and easily deployable.

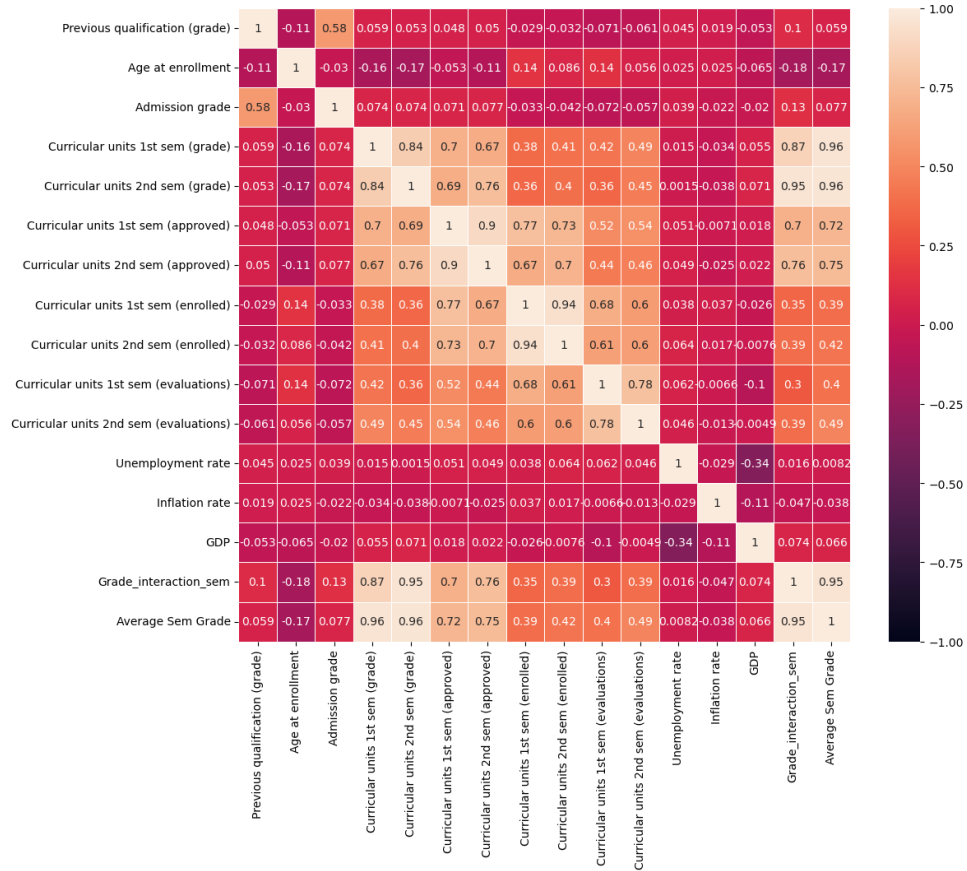
While our model demonstrated strong predictive performance, it is not without limitations. Future work should incorporate additional behavioral and psychosocial indicators—such as attendance patterns, motivation, and mental health—and consider more advanced modeling techniques using interaction effects and time-series data. Continuous iteration and institutional feedback will be crucial in evolving this framework into a reliable early-warning system.

In conclusion, this project demonstrates the powerful intersection of machine learning and educational insight. By identifying actionable predictors and offering practical recommendations, our approach equips higher education institutions with a data-driven toolkit to reduce dropout rates, support student success, and promote educational equity.

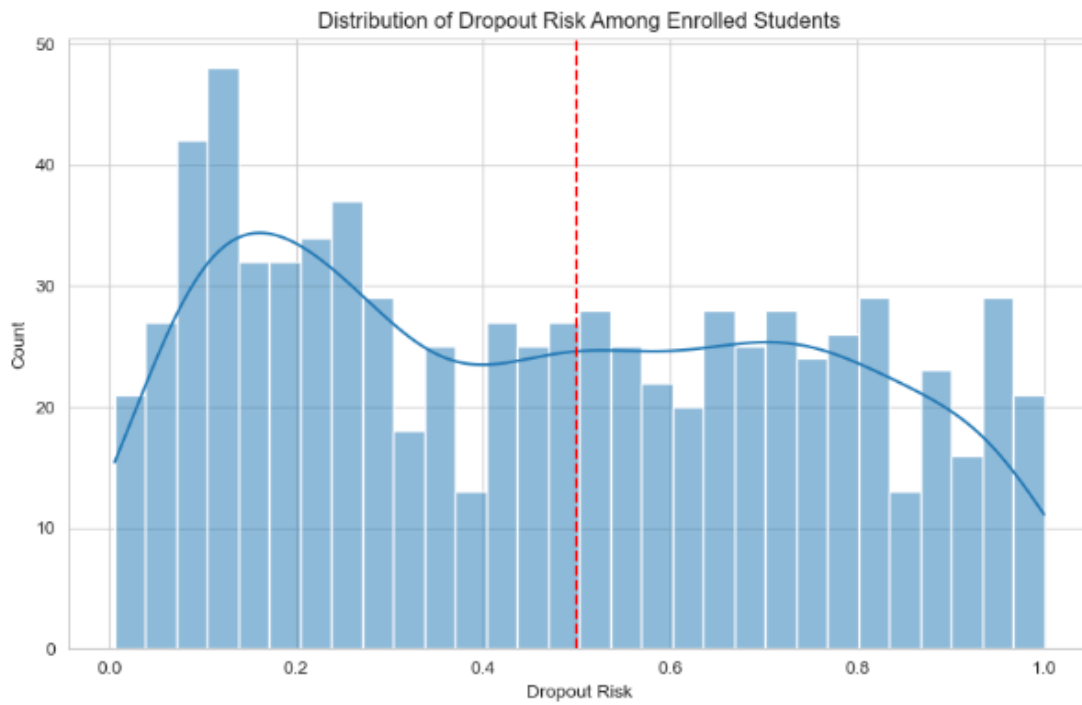
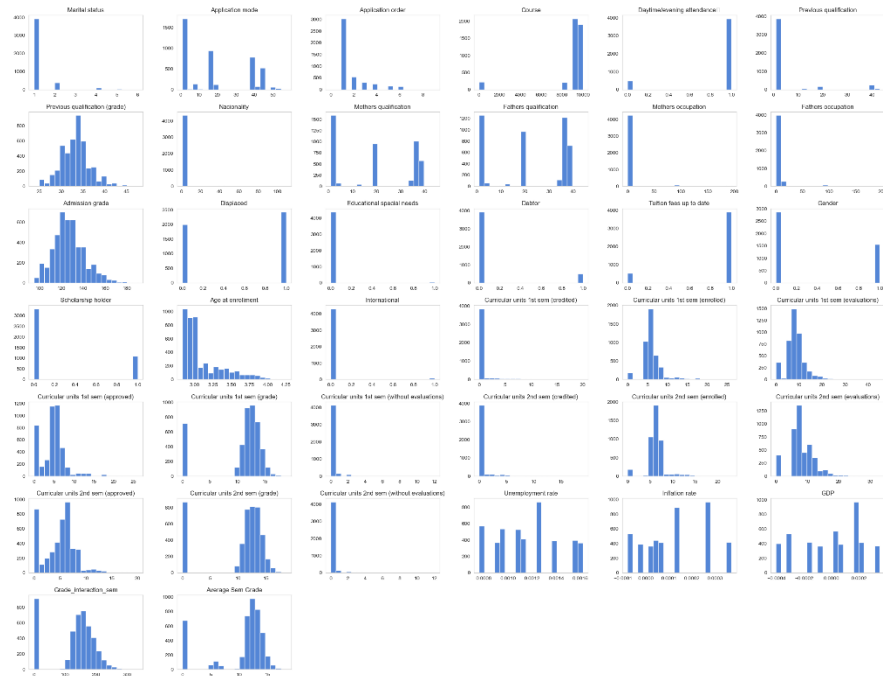
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Attachments

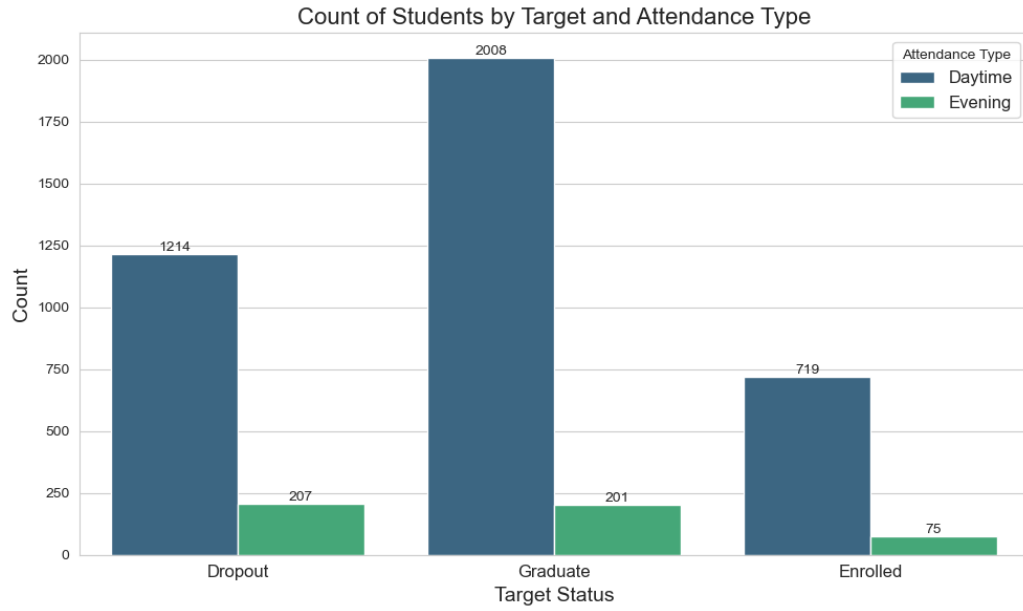
- The Discovery Phase



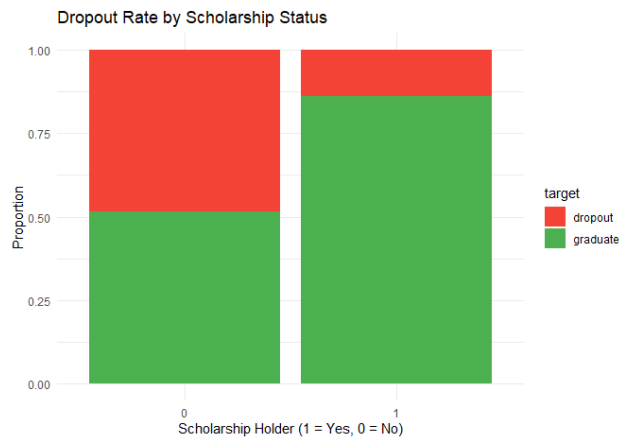
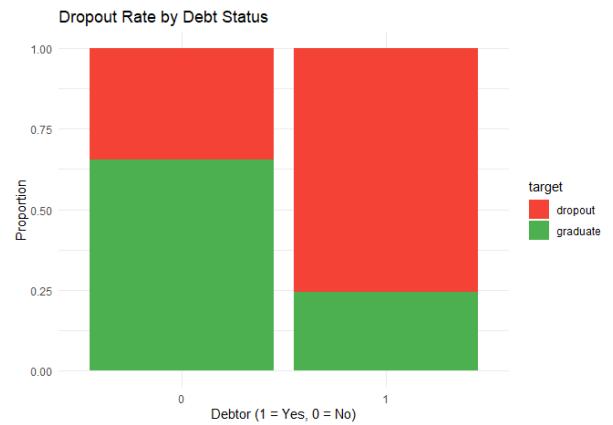
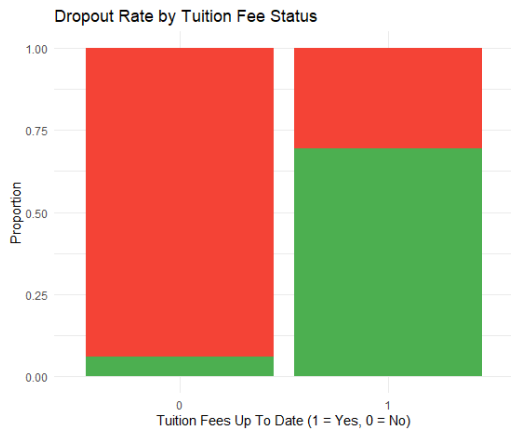
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- Financial Status Dimension Visuals**



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#Financial Status

Gini Score for Random Forest

```
> table(train_fin$target)
```

```
dropout graduate
995    2102
```

```
>
```

```
> rf_fin_model <- randomForest(target ~ ., data = train_fin, importance = TRUE)
```

```
>
```

```
> gini_scores <- data.frame(
```

```
+ Feature = rownames(importance(rf_fin_model)),
```

```
+ MeanDecreaseGini = importance(rf_fin_model)[, "MeanDecreaseGini"]
```

```
+ )
```

```
>
```

```
> print(gini_scores)
```

	Feature	MeanDecreaseGini
debtor	debtor	24.63736
tuitionfeesuptodate	tuitionfeesuptodate	132.92422
scholarshipholder	scholarshipholder	48.87456

Logistic regression on financial features only

```
fin_model <- glm(target ~ ., data = fin_data, family = "binomial")
```

```
summary(fin_model)
```

Call:

```
glm(formula = target ~ ., family = "binomial", data = fin_data)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-2.1293	-0.9858	0.4678	0.9687	2.6923

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.6150	0.2011	-13.005	< 2e-16 ***
debtor	-0.9821	0.1515	-6.482	9.04e-11 ***
tuition_fees_up_to_date	3.1280	0.2023	15.464	< 2e-16 ***
scholarship_holder	1.6445	0.1108	14.846	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 4859.8 on 3629 degrees of freedom

Residual deviance: 3777.4 on 3626 degrees of freedom

AIC: 3785.4

Number of Fisher Scoring iterations: 5

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#ROC-AUC Scores

```
> rf_fin_model <- randomForest(target ~ ., data = train_fin, importance = TRUE)
>
> # Predict probabilities for dropout class
> rf_probs <- predict(rf_fin_model, newdata = test_fin, type = "prob")[, "dropout"]
>
> # Calculate AUC
> library(pROC)
> roc_obj <- roc(test_fin$target, rf_probs, levels = c("graduate", "dropout"), direction = "<")
> auc(roc_obj)
Area under the curve: 0.6554
>
> log_fin_model <- glm(target ~ ., data = train_fin, family = "binomial")
>
> log_probs <- predict(log_fin_model, newdata = test_fin, type = "response")
>
> roc_log <- roc(test_fin$target, log_probs, levels = c("graduate", "dropout"), direction = "<")
> auc(roc_log)
Area under the curve: 0.7214
```

#Accuracy Score

```
#Train model
> rf_fin_model <- randomForest(target ~ ., data = train_fin)
> # Predict on test
> rf_fin_pred <- predict(rf_fin_model, newdata = test_fin)
> # Accuracy
> rf_fin_acc <- mean(rf_fin_pred == test_fin$target)
> cat("???? Random Forest Accuracy (Financial Features):", round(rf_fin_acc * 100, 2), "%\n")
???? Random Forest Accuracy (Financial Features): 60.62 %

library(nnet)
>
> # Train model
> mlogit_fin_model <- multinom(target ~ ., data = train_fin)
# weights: 15 (8 variable)
initial value 3401.303646
iter 10 value 2761.396181
final value 2754.515503
converged
> # Predict
> mlogit_fin_pred <- predict(mlogit_fin_model, newdata = test_fin)
> # Accuracy
> mlogit_fin_acc <- mean(mlogit_fin_pred == test_fin$target)
> cat("???? Multinomial Logistic Accuracy (Financial Features):", round(mlogit_fin_acc * 100,
2), "%\n")
???? Multinomial Logistic Accuracy (Financial Features): 60.62 %
```

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- Grade Progression Dimension Visuals

POOR PERFORMERS - FINAL MODEL

Random Forest Model Specifications:

```
Call:
  randomForest(x = improved$train_clean[selected_features], y = improved$train_clean$Target, ntree = 500, importance = TRUE)
  Type of random forest: classification
    Number of trees: 500
  No. of variables tried at each split: 4

  OOB estimate of error rate: 9.35%
Confusion matrix:
      Dropout Graduate class.error
Dropout      820      64      0.0724
Graduate      55     334      0.1414
```

Performance Metrics:

- **Accuracy:** 85.8%
- **Sensitivity:** 75.26%
- **Specificity:** 90.45%
- **Features Used:** 20 out of 41 (51.2% reduction)
- **Training Samples:** 1273
- **Outliers Removed:** 0

Confusion Matrix:

	Predicted Dropout	Predicted Graduate
Actual Dropout	199	24
Actual Graduate	21	73

Top 5 Important Features:

Feature	MeanDecreaseGini
Curricular.units.2nd.sem..approved.	97.30
Curricular.units.1st.sem..approved.	54.36
Curricular_units_2nd_sem_grade_reflect_log	43.11
Curricular.units.2nd.sem..grade.	39.56
Tuition.fees.up.to.date	28.90

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AVERAGE PERFORMERS - FINAL MODEL

Random Forest Model Specifications:

```
Call:
  randomForest(x = improved$train_clean[selected_features], y = improved$train_clean$Target,      ntree = 500, importance = TRUE)
      Type of random forest: classification
      Number of trees: 500
No. of variables tried at each split: 3

OOB estimate of error rate: 10.69%
Confusion matrix:
      Dropout Graduate class.error
Dropout    91      90    0.49724
Graduate    17     803    0.02073
```

Performance Metrics:

- **Accuracy:** 90%
- **Sensitivity:** 98.05%
- **Specificity:** 53.33%
- **Features Used:** 15 out of 41 (63.4% reduction)
- **Training Samples:** 1001
- **Outliers Removed:** 0

Confusion Matrix:

	Predicted Dropout	Predicted Graduate
Actual Dropout	24	4
Actual Graduate	21	201

Top 5 Important Features:

Feature	MeanDecreaseGini
Curricular.units.2nd.sem..approved.	54.47
Curricular.units.1st.sem..approved.	27.09
Tuition.fees.up.to.date	24.19
Curricular.units.1st.sem..grade.	23.71
Curricular_units_1st_sem_grade_reflect_log	23.51

GOOD PERFORMERS - FINAL MODEL

Random Forest Model Specifications:

```
Call:
  randomForest(x = improved$train_clean[selected_features], y = improved$train_clean$Target,      ntree = 500, importance = TRUE)
      Type of random forest: classification
      Number of trees: 500
No. of variables tried at each split: 3

OOB estimate of error rate: 8.39%
Confusion matrix:
      Dropout Graduate class.error
Dropout    26      47    0.64384
Graduate     6     553    0.01073
```

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Performance Metrics:

- **Accuracy:** 89.81%
- **Sensitivity:** 97.12%
- **Specificity:** 33.33%
- **Features Used:** 15 out of 41 (63.4% reduction)
- **Training Samples:** 632
- **Outliers Removed:** 0

Confusion Matrix:

	Predicted Dropout	Predicted Graduate
Actual Dropout	6	4
Actual Graduate	12	135

Top 5 Important Features:

Feature	MeanDecreaseGini
Curricular.units.2nd.sem..approved.	23.50
Curricular_units_1st_sem_grade_reflect_log	12.20
Curricular.units.1st.sem..grade.	11.97
Curricular.units.1st.sem..approved.	11.70
Curricular_units_2nd_sem_grade_reflect_log	10.69

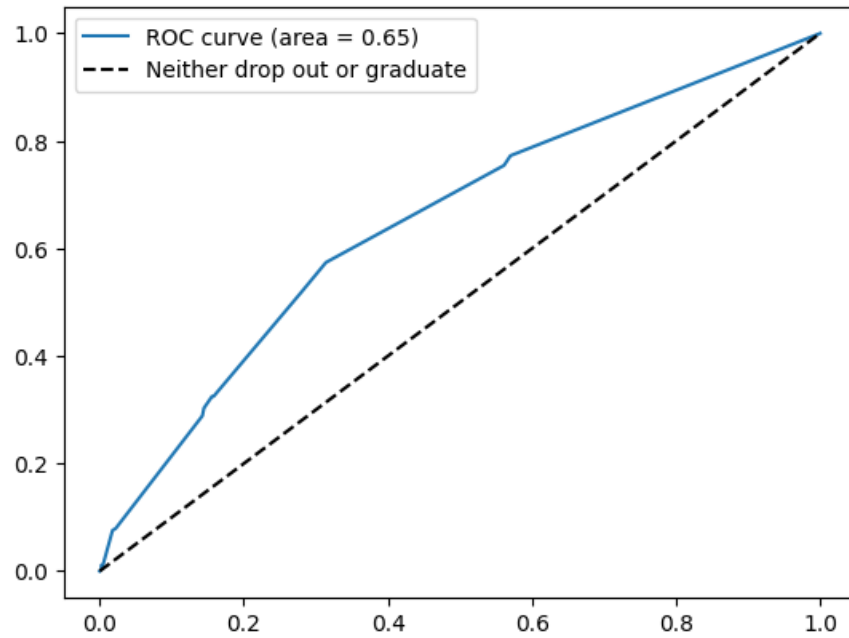
- **Demographics Dimension Visuals**

```
1st: Logistic Regression Model
0.6517906336088154
      precision    recall  f1-score   support

     0       0.70      0.74      0.72      2209
     1       0.56      0.52      0.54      1421

 accuracy          0.65          3630
 macro avg         0.63          0.63          0.63          3630
 weighted avg      0.65          0.65          0.65          3630
```

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```
[(np.float64(0.2673592554757338), 'Marital_status'), (np.float64(1.0624482982978878), 'Gender'), (np.float64(-0.35178260406811984), 'Displaced')]  
[-0.94383335]
```

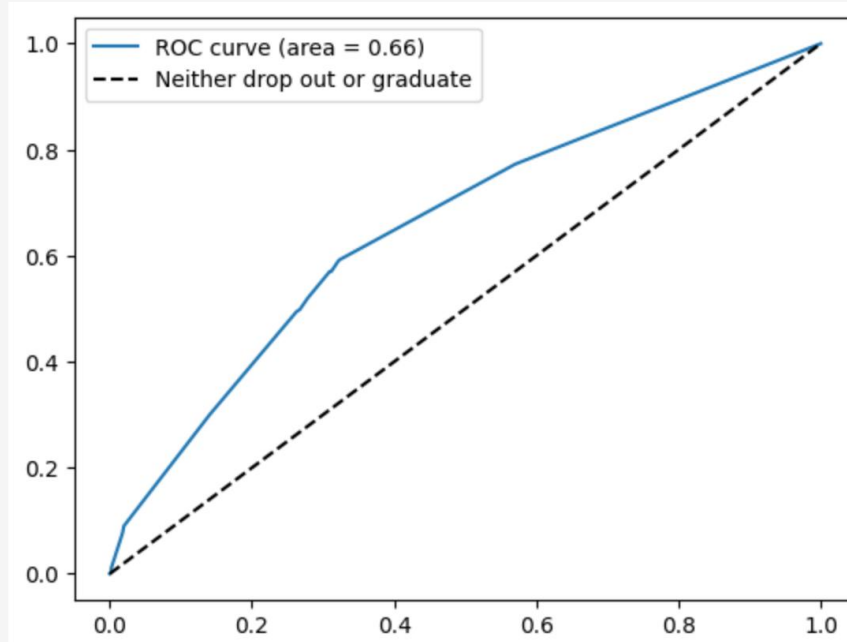
	Feature	Importance
1	Gender	0.608945
2	Displaced	0.211642
0	Marital_status	0.179412

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2nd: Random Forest Model

0.6506887052341598

	precision	recall	f1-score	support
0	0.66	0.87	0.75	2209
1	0.61	0.30	0.41	1421
accuracy			0.65	3630
macro avg	0.63	0.59	0.58	3630
weighted avg	0.64	0.65	0.62	3630



	Graduated	Dropped out	Dropout Rate (%)
Gender			
0	1661	720	30.239395
1	548	701	56.124900
Married			
Marital_status			
1	2015	1184	37.011566
2	148	179	54.740061
3	1	1	50.000000
4	33	42	56.000000
5	11	11	50.000000
6	1	4	80.000000
Displaced			
Displaced			
0	885	752	45.937691
1	1324	669	33.567486

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- Final Model (Logistic Regression) Daytime

```
Accuracy: 0.9065166795965865
```

	precision	recall	f1-score	support
Dropout	0.91	0.84	0.87	964
Graduate	0.91	0.95	0.93	1614
accuracy			0.91	2578
macro avg	0.91	0.89	0.90	2578
weighted avg	0.91	0.91	0.91	2578

```
[[ 806 158]
 [ 83 1531]]
```

Top 10 most important features:

	Feature	Importance
0	Tuition fees up to date	2.552562
1	Curricular units 2nd sem (approved)	1.143167
2	Debtor	0.867942
3	Curricular units 1st sem (without evaluations)	0.621023
4	Displaced	0.589029
5	Curricular units 2nd sem (enrolled)	0.582333
6	Curricular units 1st sem (approved)	0.473546
7	Scholarship holder	0.439836
8	Curricular units 1st sem (enrolled)	0.419208
9	Curricular units 2nd sem (credited)	0.415071

- Final Model (Logistic Regression) Evening Students

```
Accuracy: 0.8975950349107835
```

	precision	recall	f1-score	support
Dropout	0.89	0.83	0.86	964
Graduate	0.90	0.94	0.92	1614
accuracy			0.90	2578
macro avg	0.90	0.88	0.89	2578
weighted avg	0.90	0.90	0.90	2578

```
[[ 801 163]
 [ 101 1513]]
```

Top 5 most important features:

	Feature	Importance
0	Curricular units 2nd sem (approved)	0.257294
1	Curricular units 1st sem (approved)	0.151934
2	Curricular units 1st sem (grade)	0.095976
3	Tuition fees up to date	0.049692
4	Age at enrollment	0.044694

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- Final Model (Random Forest) Educational Special Needs

```
Accuracy: 0.78125

              precision    recall  f1-score   support

   Dropout      0.82      0.64      0.72      14
   Graduate      0.76      0.89      0.82      18

 accuracy      0.78      0.78      0.78      32
 macro avg      0.79      0.77      0.77      32
 weighted avg    0.79      0.78      0.78      32

[[ 9  5]
 [ 2 16]]

Top 10 most important features:
              Feature  Importance
0  Curricular units 2nd sem (approved)  0.166015
1  Curricular units 1st sem (grade)    0.087884
2  Curricular units 2nd sem (grade)    0.086190
3  Curricular units 1st sem (approved)  0.082566
4  Admission grade                     0.078754
5  Mothers occupation                   0.078678
6  Grade_interaction_sem               0.066737
7  Previous qualification (grade)      0.057221
8  Application order                  0.040689
9  Fathers qualification               0.038301
```

- Overall Model

```
Accuracy: 0.9121900826446281

              precision    recall  f1-score   support

   Dropout      0.92      0.85      0.88     1134
   Graduate      0.91      0.95      0.93     1770

 accuracy      0.91      0.91      0.91     2904
 macro avg      0.91      0.90      0.91     2904
 weighted avg    0.91      0.91      0.91     2904

[[ 966 168]
 [  87 1683]]

Top 10 most important features:
              Feature  Importance
0  Tuition fees up to date  2.150084
1  International           1.394879
2  Debtor                  1.016924
3  Curricular units 2nd sem (approved)  0.920284
4  Curricular units 2nd sem (enrolled)  0.899100
5  Scholarship holder      0.800806
6  Gender                  0.664180
7  Curricular units 1st sem (approved)  0.618386
8  Age at enrollment       0.401639
9  Curricular units 2nd sem (without evaluations)  0.391535
```

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